Computer Programming 1 - Exam 2019

Question 1 [20 Marks]

Write a Java program to print the following pattern using loops.

1 1 1 1 1

2 2 2 2

3 3 3

4 4

5

Question 2 [20 Marks]

It is given that the array A has a fixed size of 10 integers. Write a java program to:

1. Read the following elements one by one from the user via console and assign them in A. [6 marks]

   A = [3, 3, 4, 5, 5, 5, 7, 7, 1, 1]

2. Find the most occurring element in this array. [7 marks]

3. Find the first index of the element which has the maximum value. In addition, print its value. [7 marks]

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Question 3 [60 Marks]

Create a class called **Invoice** that a hardware store might use to represent an invoice for an item sold at the store.

- An Invoice should include four pieces of information as instance variables: **a part number** (type String), **a part description** (type String), **a quantity of the item being purchased** (type int) and a **price per item** (double). [10 marks]
- Your class should have a constructor that initializes the four instance variables. If the quantity is not positive, it should be set to 0. If the price per item is not positive, it should be set to 0.0. [10 marks]
- Provide a set and a get method for each instance variable. [10 marks]
- In addition, provide a method named **getInvoiceAmount()** that calculates the invoice amount (i.e., multiplies the quantity by the price per item), then returns the amount as a double value. [10 marks]
- Write a **printInvoice()** method to print the details such as part number, description, the quantity, price per item, and the total price. [10 marks]
- Write a test application named **InvoiceTest** to create two instances of the Invoice class and print the details of each. [10 marks]